

Wavefronts in Biased Population Dynamics

joint work with Andrea Corli (Ferrara, IT) and Luisa Malaguti (Modena & Reggio Emilia, IT)

Diego Berti
(University of Turin, Italy)



UNIVERSITÀ
DI TORINO

Equadiff 2024

Diffusion processes in population dynamics and biology

13 June 2024, Karlstad University

The equation

We deal with the following equation:

$$u_t + f(u)_x = \{D(u) u_x\}_x + g(u) \quad \text{for } x \in \mathbb{R}, t \geq 0. \quad (1)$$

Here, the unknown is

$$u = u(x, t) \in [0, 1],$$

while f , D and g are given and such that

- **diffusivity** $D \in C^1$ is possibly degenerate ($D = 0$ somewhere) or even negative in some interval (**mixed forward-backward equation**);
- **source** $g \in C^0$ vanishes at some isolated points (equilibria);
- **convection** $f \in C^1$. (For convenience, $f(0) = 0$.)

Our focus: Traveling waves

Traveling wave solutions (Aronson-Weinberger (1978), Gilding-Kersner (2004)) are solutions u which take the form

$$u(x, t) = \varphi(x - ct),$$

for some wave profile $\varphi = \varphi(\xi)$ and wave speed $c \in \mathbb{R}$.

We then formally rephrase (1) as

$$(D(\varphi) \varphi')' + (c - f'(\varphi)) \varphi' + g(\varphi) = 0. \quad (2)$$

TW of (1) if there exist φ , differentiable a.e., and c such that φ satisfies (2) in distributional sense.

A bounded TW, whose profile is defined globally (on $\mathbb{R}$), non-constant and **monotone**, and

$$\varphi(-\infty) = \ell^+ \quad \varphi(+\infty) = \ell^-,$$

is said to be a **wavefront** (WF) from ℓ^+ to ℓ^- , where $g(\ell^\pm) = 0$.

$\mathcal{J}$ is the set of **admissible speeds**, that is the set of $c \in \mathbb{R}$ such that the considered problem admits a WF with speed c .

Main goal is to study:

– **Existence, uniqueness (non-uniqueness) and regularity** (where D degenerates) of WFs, **highlighting how the convection f affects these results**

Remark: Uniqueness is always meant up to space shifts.

Key issue is also:

– Information on $\mathcal{J}$.

A bounded TW, whose profile is defined globally (on $\mathbb{R}$), non-constant and **monotone**, and

$$\varphi(-\infty) = \ell^+ \quad \varphi(+\infty) = \ell^-,$$

is said to be a **wavefront** (WF) from ℓ^+ to ℓ^- , where $g(\ell^\pm) = 0$.

$\mathcal{J}$ is the set of **admissible speeds**, that is the set of $c \in \mathbb{R}$ such that the considered problem admits a WF with speed c .

Main goal is to study:

– **Existence, uniqueness (non-uniqueness) and regularity** (where D degenerates) of WFs, **highlighting how the convection f affects these results**

Remark: Uniqueness is always meant up to space shifts.

Key issue is also:

– Information on $\mathcal{J}$.

Classical results:

in the case $D > 0$:

-) Fisher (1937)
-) Kolmogorov-Petrovskii-Piskunov (1937)
-) Aronson-Weinberger (1978)
-) Fife (1979)

D is **negative** somewhere:

Few general results (equation is generally ill-posed)

Plotnikov (1993–1996), Evans-Portilheiro (2004) (viscous regularization)

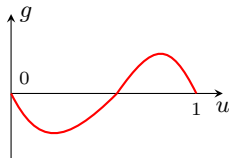
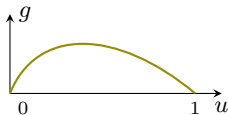
Bertsch, Mascia, Smarrazzo, Terracina, Tesei (2007–).

Previous results to Equations similar to (1): Maini-Malaguti-Marcelli-Matucci (2006-07), Ruggerini-Kuzmin (2011), Ferracuti-Marcelli-Papalini (2009), Bao-Zhou (2014-17)

Always with $f = 0$.

The source term g

In this talk: $g \in C^0([0, 1])$, $g(0) = g(1) = 0$ and either g is **monostable** or **bistable**:



We consider WFs s.t.

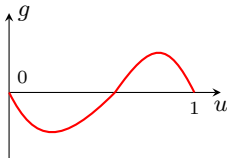
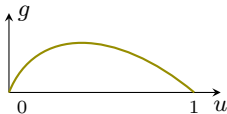
$$\varphi(-\infty) = 1, \quad \varphi(\infty) = 0$$

Typical results in the case $f = 0$

-) if g is **monostable**, WFs if and only if $c \geq c^* = c^*(Dg)$, and the corresponding profile is unique (up to space shifts);
-) if g is **bistable**, WFs for **at most** a unique value of wave speed c^* , and the corresponding profile is unique (up to space shifts).
-) Moreover, profiles are smooth except (possibly) at $0, 1$.

The source term g

In this talk: $g \in C^0([0, 1])$, $g(0) = g(1) = 0$ and either g is **monostable** or **bistable**:



We consider WFs s.t.

$$\varphi(-\infty) = 1, \quad \varphi(\infty) = 0$$

Typical results in the case $f = 0$

-) if g is **monostable**, WFs if and only if $c \geq c^* = c^*(Dg)$, and the corresponding profile is unique (up to space shifts);
-) if g is **bistable**, WFs for **at most** a unique value of wave speed c^* , and the corresponding profile is unique (up to space shifts).
-) Moreover, profiles are smooth except (possibly) at 0, 1.

A first look at the influence of convection: a shift of $\mathcal{J}$

Take $D \geq 0$ and g monostable

B., Corli, Malaguti (2020) (see also In Marcelli, Malaguti (2004) and Marcelli, Papalini (2018))

WFs exist for $\mathcal{J} = [c^*, +\infty)$, where $c^* = c^*(Dg, f)$ is such that

$$\sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi} \leq c^* \leq 2 \sqrt{\sup_{\varphi \in (0,1]} \frac{D(\varphi)g(\varphi)}{\varphi}} + \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi}.$$

that reads

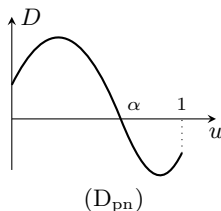
$$c_f \leq c^* \leq c_{Dg} + c_f$$

-) For simplicity, $(Dg)'(0) = 0$.

-) c_f comes from the **diffusion-convection** equation of the type $u_t + f(u)_x = u_{xx}$.

A second consequence of f : occurrence of no C^1 solutions

The case g monostable and (D_{pn})



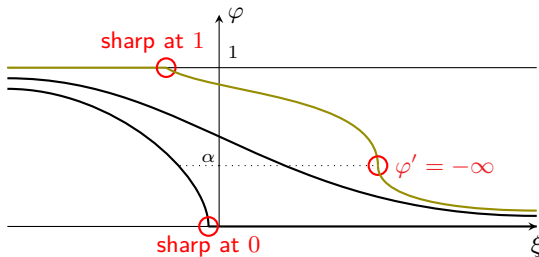
Theorem [B.-Corli-Malaguti (2021)]

- $\mathcal{J} = [c_{pn}^*, +\infty)$, with estimates on c_{pn}^* depending on Dg and f , that are coherent with the case $f = 0$;
- For every admissible speed, there is a **unique** wavefront solution (up to space shifts);
- precise estimates on c_{pn}^{*1} ;

¹[B., Corli, Malaguti (2020)]

Sharp profiles

- at 0 or 1 when $c = c_{pn}^*$; at α also for $c > c_{pn}^*$;
- f nonlinear $\Rightarrow$ the occurrence of *infinity-slope profiles*

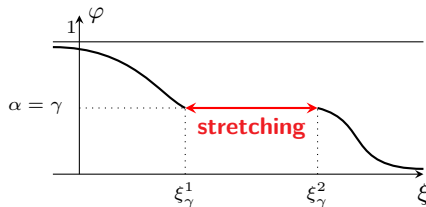
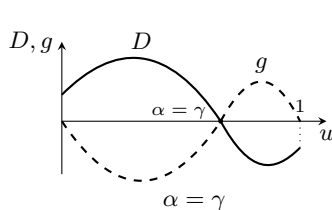


To have a vertical profile f needs to change its convexity, from convex to concave:
roughly speaking, the behavior of f strongly contrasts that of D .

A third surprising effect of convection: existence and nonuniqueness of solutions

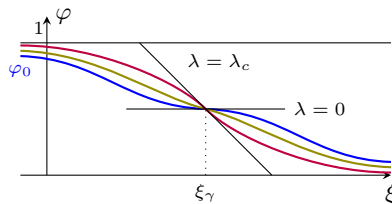
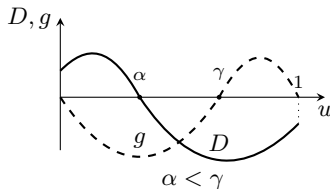
Theorem [B.-Corli-Malaguti (2022)]:

- under sufficient conditions: $\mathcal{J} = [c_1^*, c_2^*]$;
- for $c \in \mathcal{J}$, the profile is unique up to shifts and **stretchings** at level γ .



Theorem [B., Corli, Malaguti (2022)]:

- under sufficient conditions: $\mathcal{J} = [c_3^*, c_4^*]$;
- For $c \in \text{int}\mathcal{J}$, there is a **family of profiles** $\{\varphi_\lambda\}_\lambda$, determined by the **slope** at ξ_γ ;
- φ_0 **can be stretched** if g is superlinear at γ .



Remark: precise estimates on $\inf \mathcal{J}$ and $\sup \mathcal{J}$ can be given.

Theorem 1 (Sufficient conditions [B.-Corli-Malaguti 2022])

Assume that f is strictly concave. Then, WFs if

(a) $\alpha > \gamma$, $\dot{g}(\gamma) > 0$ and

$$f'(\gamma) - \frac{f(1) - f(\alpha)}{1 - \alpha} \geq 2 \left(\sqrt{\sup_{[\alpha, 1)} \delta(Dg, 1)} + \sqrt{\sup_{[0, \gamma)} \delta(Dg, \gamma)} \right); \quad (3)$$

(b) $\alpha = \gamma$, and (3);

(c) $\alpha < \gamma$ and

$$f'(\gamma) - \frac{f(1) - f(\gamma)}{1 - \gamma} > 2 \left(\sqrt{\sup_{[\gamma, 1)} \delta(Dg, 1)} + \sqrt{\sup_{[0, \gamma] \setminus \{\alpha\}} \delta(Dg, \alpha)} \right).$$

Notation: $\delta(Dg, 1)$ is the *difference quotient* of the function Dg w.r.t. a point 1:

$$\delta(Dg, 1)(s) := \frac{(Dg)(1) - (Dg)(s)}{1 - s} = \frac{D(s)g(s)}{s - 1} > 0.$$

Main tool: reduction to singular first-order problems

[–] **lack of regularity** $\implies$ **not** possible to apply a standard dynamical system analysis

[–] method well-formalized by **Marcelli-Malaguti** in the early '00s: **reduction** (in regions where $D \neq 0$) into the following singular first-order boundary problem

$$\begin{cases} \dot{z}(\varphi) = f'(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \text{for } \varphi \in (0, \alpha) \cup (\alpha, 1), \\ z(\varphi) < 0 & \text{for } \varphi \in (0, \alpha), \\ z(\varphi) > 0 & \text{for } \varphi \in (\alpha, 1), \\ z(0) = z(\alpha) = z(1) = 0. \end{cases}$$

with (formally):

$$z(\varphi) := D(\varphi)\varphi',$$

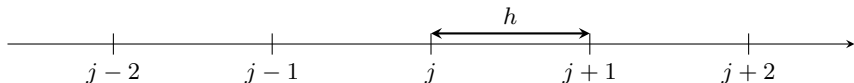
where the r.h.s. is evaluated at $\varphi^{-1}(\varphi)$.

[–] **refined use of upper- and lower-solutions** based on **comparison-type techniques**.

Applications:
a biological model with biased movements

-) *lattice-based stochastic model*, (from [Johnston et al \(2017\)](#)):

- agents are split in (non-interacting) isolated and grouped individuals;
- both classes can reproduce, die and move, with different rates;
- discrete model both in space $h > 0$ and time $\tau > 0$;



- P_m^i, P_m^g : "local" movement probabilities of (isolated and grouped) individuals;
- P_b^i, P_b^g : probabilities of birth;
- P_d^i, P_d^g : probabilities of death;
- c_j : probability of occupancy of the j -th site ($1 - c_j$: probability that j is vacant).

New features in [B.-Corli-Malaguti \(2023\)](#):

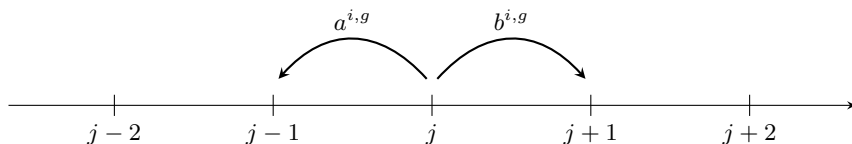
Parameters $a^i, b^i \geq 0$ and $a^g, b^g \geq 0$

characterizing a **biased movement** (to the left and to the right). It could be expressed by probabilities.

-) Motivations: search for food, migration, escape from predators, ...

-) For example, isolated individuals move

- to the left if $a^i - b^i > 0$;
- to the right if $a^i - b^i < 0$.



The variation δc_j of c_j during a time-step $\tau > 0$ is

$$\begin{aligned} \delta c_j = & \frac{P_m^i}{2} \left[b^i c_{j-1}(1 - c_j)(1 - c_{j-2}) + a^i c_{j+1}(1 - c_j)(1 - c_{j+2}) \right. \\ & \left. - (a^i + b^i) c_j(1 - c_{j-1})(1 - c_{j+1}) \right] \\ & + \frac{P_m^g}{2} \left[a^g c_{j-1}(1 - c_j) + b^g c_{j+1}(1 - c_j) - a^g c_j(1 - c_{j+1}) - b^g c_j(1 - c_{j-1}) \right] \\ & - \frac{P_m^g}{2} \left[a^g c_{j-1}(1 - c_j)(1 - c_{j-2}) + b^g c_{j+1}(1 - c_j)(1 - c_{j+2}) \right. \\ & \left. - (a^g + b^g) c_j(1 - c_{j-1})(1 - c_{j+1}) \right] + \text{reaction terms}, \end{aligned}$$

where reaction terms =

$$\begin{aligned} = & \frac{P_b^i}{2} [c_{j-1}(1 - c_j)(1 - c_{j-2}) + c_{j+1}(1 - c_j)(1 - c_{j+2})] \\ & + \frac{P_b^g}{2} [c_{j-1}(1 - c_j) + c_{j+1}(1 - c_j)] \\ & - \frac{P_b^g}{2} [c_{j-1}(1 - c_j)(1 - c_{j-2}) + c_{j+1}(1 - c_j)(1 - c_{j+2})] - \frac{P_d^i}{2} [c_j(1 - c_{j-1})(1 - c_{j+1})] \\ & - \frac{P_d^g}{2} [c_j] + \frac{P_d^g}{2} [c_j(1 - c_{j-1})(1 - c_{j+1})]. \end{aligned}$$

Rem: novelty w.r.t. Johnston et al ('17): $a^{i,g}$ and $b^{i,g}$

Toward the continuum model

The continuum model is obtained as follows:

- replace c_j with a **smooth function** $c = c(x, t)$ and expand c around $x = jh$ at second order;
- divide by τ and **pass to the limit for $h, \tau \rightarrow 0$** while keeping h^2/τ constant (better, converging; for simplicity, think $h^2/\tau = 1$);
- **assume** for $h \rightarrow 0, \tau \rightarrow 0$:

$$\frac{P_m^{i,g} h^2}{2\tau} \sim D_{i,g}, \quad \frac{P_b^{i,g}}{2\tau} \sim \lambda_{i,g}, \quad \frac{P_d^{i,g}}{2\tau} \sim k_{i,g}, \quad P_b^{i,g}, P_d^{i,g} = O(\tau),$$

BIASED movements $\Rightarrow$ **CONVECTION** term in the continuum limit

$$a^{i,g}(\tau) \sim 1, \quad b^{i,g}(\tau) \sim 1, \quad a^{i,g}(\tau) - b^{i,g}(\tau) \sim C_{i,g} \sqrt{\tau}.$$

for some $C_{i,g} \leq 0$.

The continuum model

In conclusion we end up with (u replaces c)

$$u_t + f(u)_x = (D(u)u_x)_x + g(u)$$

with

$$f(u) = -(C_i D_i + C_g D_g) u(1-u)^2 - C_g D_g u(1-u),$$

$$D(u) = D_i (1 - 4u + 3u^2) + D_g (4u - 3u^2),$$

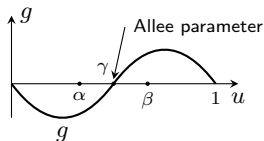
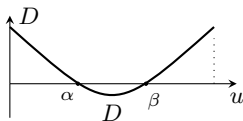
$$g(u) = \lambda_g u(1-u) + [\lambda_i - \lambda_g - (k_i - k_g)] u(1-u)^2 - k_g u.$$

Eight parameters: $C_{i,g}$, $D_{i,g}$, $k_{i,g}$ and $\lambda_{i,g}$.

- f can change convexity at most once; then, it can be:

concave, convex, convex-concave, concave-convex.

Focus on: D vanishes twice and g is bistable



- 1) D is as above iff $D_i > 4D_g > 0$;
- 2) g is bistable as above iff $k_g = 0$, $\lambda_g > 0$, $r_i := k_i - \lambda_i > 0$;
 - $\gamma \in (\alpha, \beta)$ if and only if

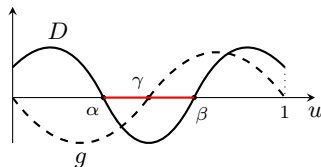
$$\frac{1 - \omega}{2 + \omega} < \frac{\lambda_g}{r_i} < \frac{1 + \omega}{2 - \omega},$$

where $\omega := \sqrt{\frac{D_i - 4D_g}{D_i - D_g}}$.

Remark:

Li, van Heijster, Marangell, Simpson (2020): Assumption 1), monostable g , and $f = 0$.
Li, van Heijster, Simpson, Wechselberger (2021) Assumptions 1) and 2), and $f = 0$

Results for D changing sign twice and g bistable



-) Kuzmin-Ruggerini (2011): ($f = 0$) no WFs if $\gamma \in (\alpha, \beta)$

Theorem [B., Corli, Malaguti (2023)]

$f \neq 0 \rightarrow \mathcal{J}$ can be an interval.

Moreover:

- There are infinitely many profiles with the same speed c .
- Explicit necessary conditions and sufficient conditions on D, g, f in order WFs exist.
- Conditions in order to have positive or negative speeds can be given.

The convection

-) A necessary condition: $C_g < 0$.
-) If f convex $\rightarrow$ No solutions;
-) Otherwise $\rightarrow$ sufficient conditions to have solutions.

The concave case

- f is strictly concave iff

$$0 \leq \frac{C_i}{|C_g|} \frac{D_i}{D_g} \leq \frac{3}{2}$$

- There exist WFs if

$$r_i(2 + \omega) + \lambda_g(1 + \omega) < \frac{8C_g^2 D_g}{81} \frac{d - 4}{(d - 1)^2}, \quad (4)$$

where $d = D_i/D_g$.

- For fixed C_g, D_g , for every $d > 4$, there exists a triangle $\mathcal{T}(d)$ in the plane (r_i, λ_g) such that (4) and the *previous* hold.

- Moreover:

[–] if $d \in (4, (5 + 2\sqrt{3})/2)$, then every $(r_i, \lambda_g) \in \mathcal{T}(d)$ provides profiles with *negative* speeds;

[–] if $d > 4 + 2\sqrt{3}$, plus another condition, then there are WFs with *positive* speeds.

Thanks for the attention!